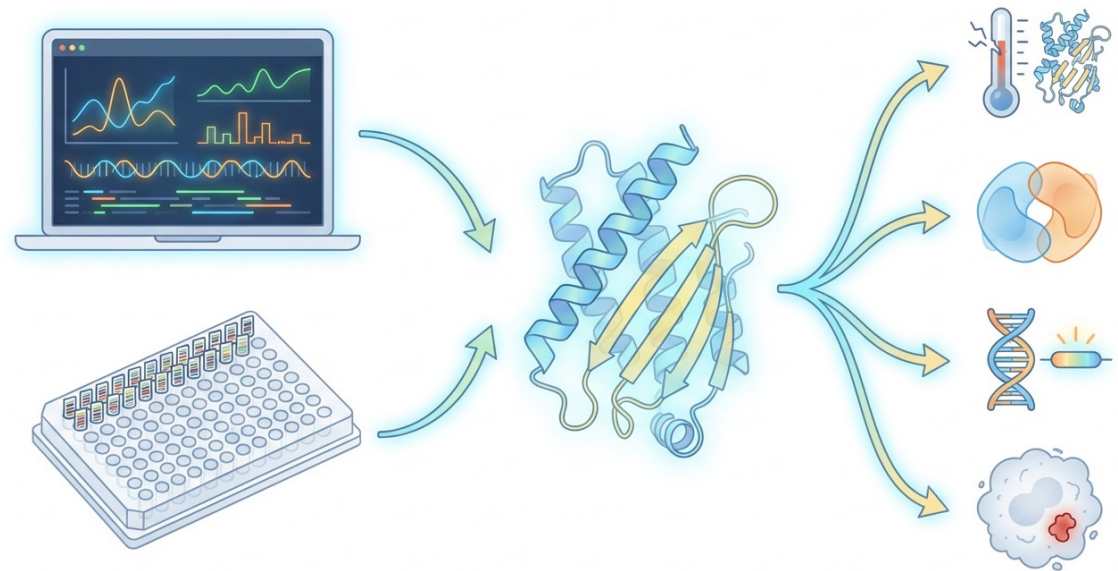


Session 4.2: Variant Effect Predictors II – VEPs, MAVEs, and molecular mechanisms

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Links and datasets

<https://sites.google.com/view/mave2025-lectures>

Why do VEPs and MAVE agree or disagree?

- Across many proteins, VEPs and MAVEs often show broad overall agreement
 - This is the idea behind using MAVEs as a truth set for benchmarking
- But when you look residue by residue, the two approaches often diverge sharply.
- For some cases, MAVEs outperform VEPs for clinically relevant variants, but in others they are rather poor at disease prediction
- We set out to better understand these difference

Why variant effect predictors and multiplexed assays agree and disagree

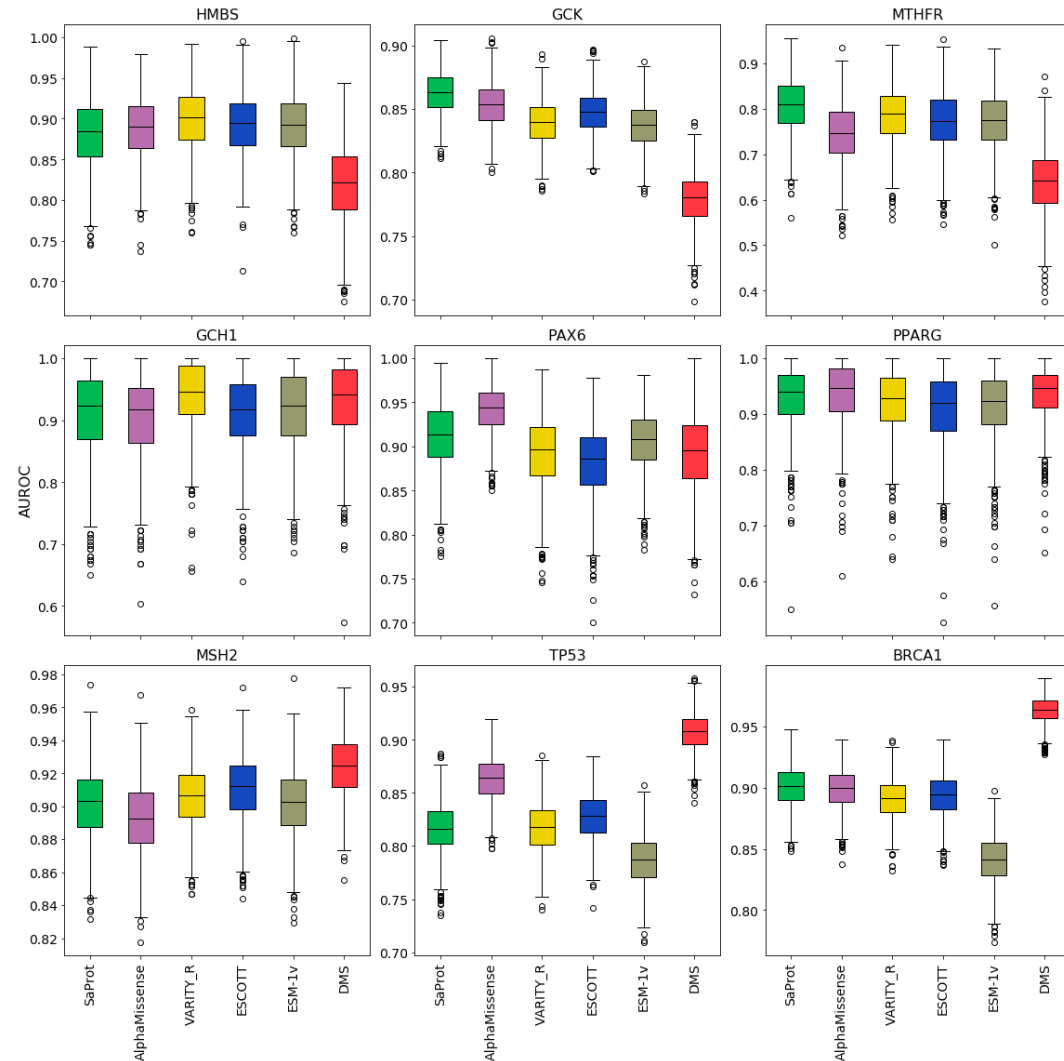
 Benjamin J. Livesey,  Joseph A. Marsh

doi: <https://doi.org/10.1101/2025.07.31.667868>

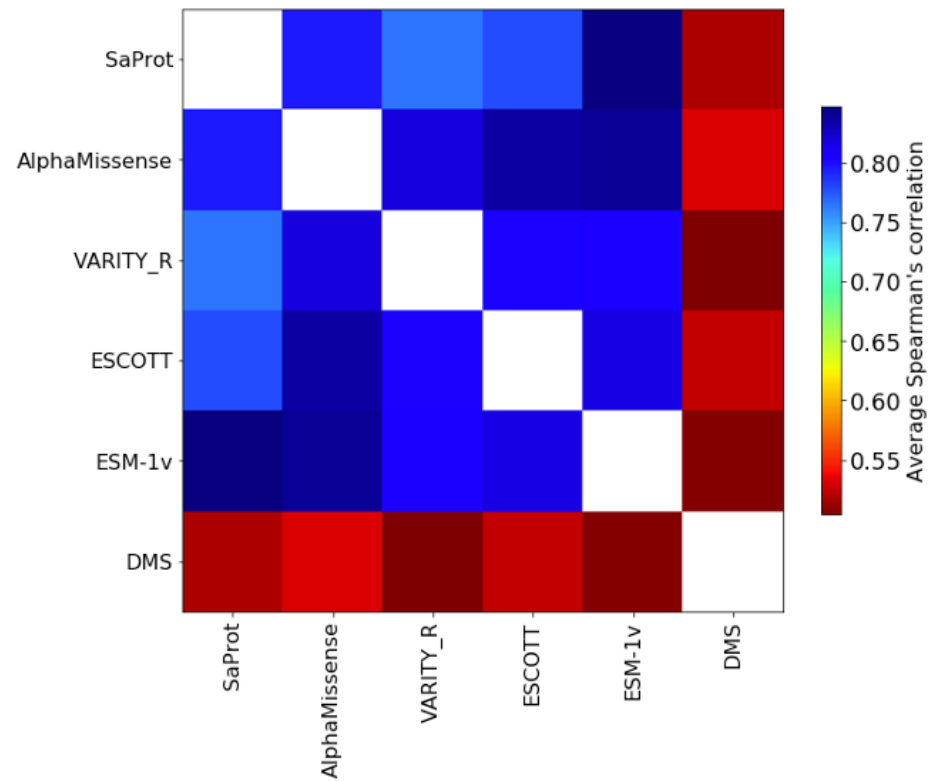
Posted August 01, 2025.

The dataset

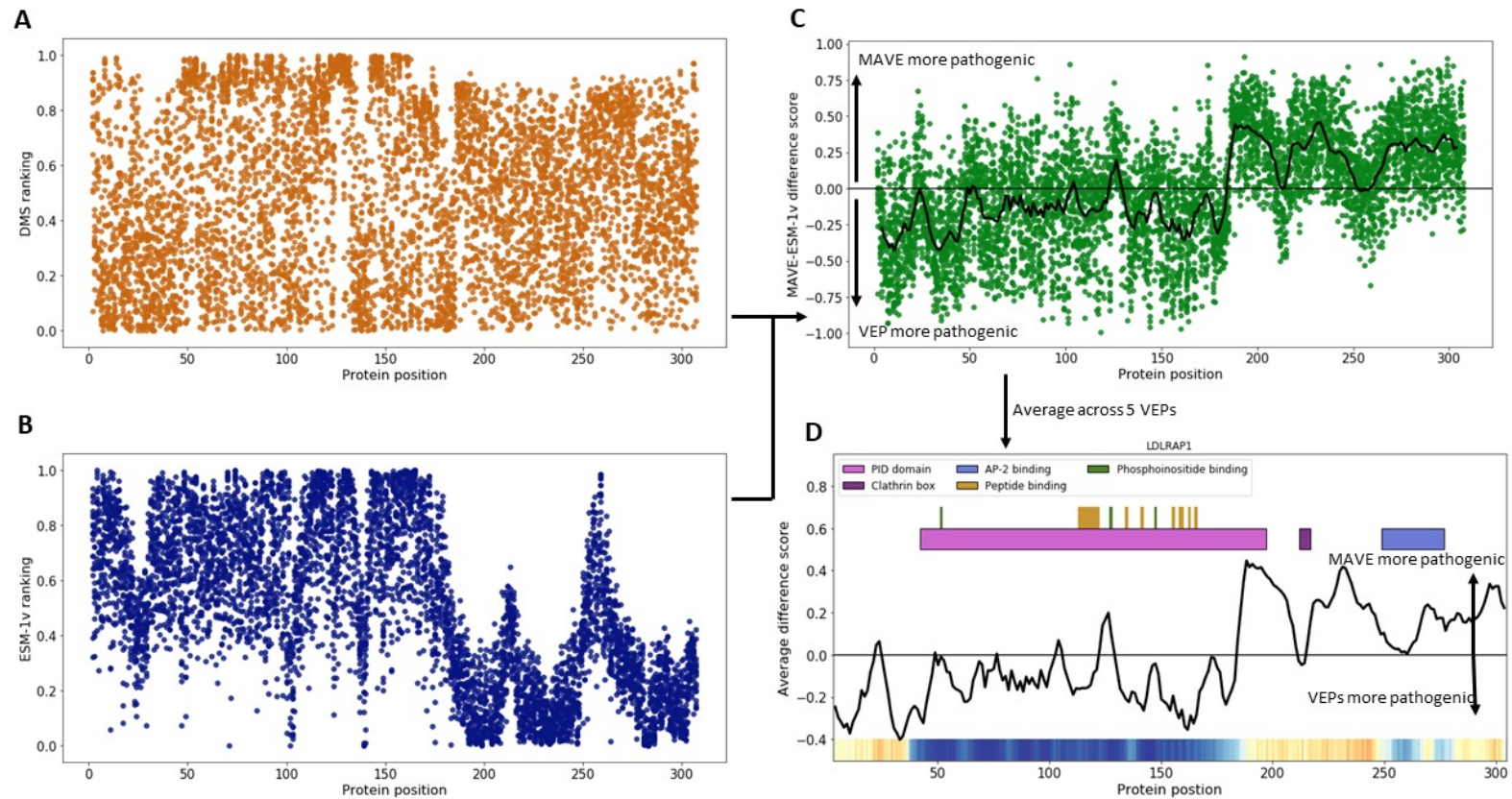
- 37 published MAVEs covering diverse proteins and assay types (yeast complementation, VAMP-seq, SGE, binding, stability).
- Five modern VEPs chosen to represent the current best methods



VEPs correlate with each other more than with MAVEs

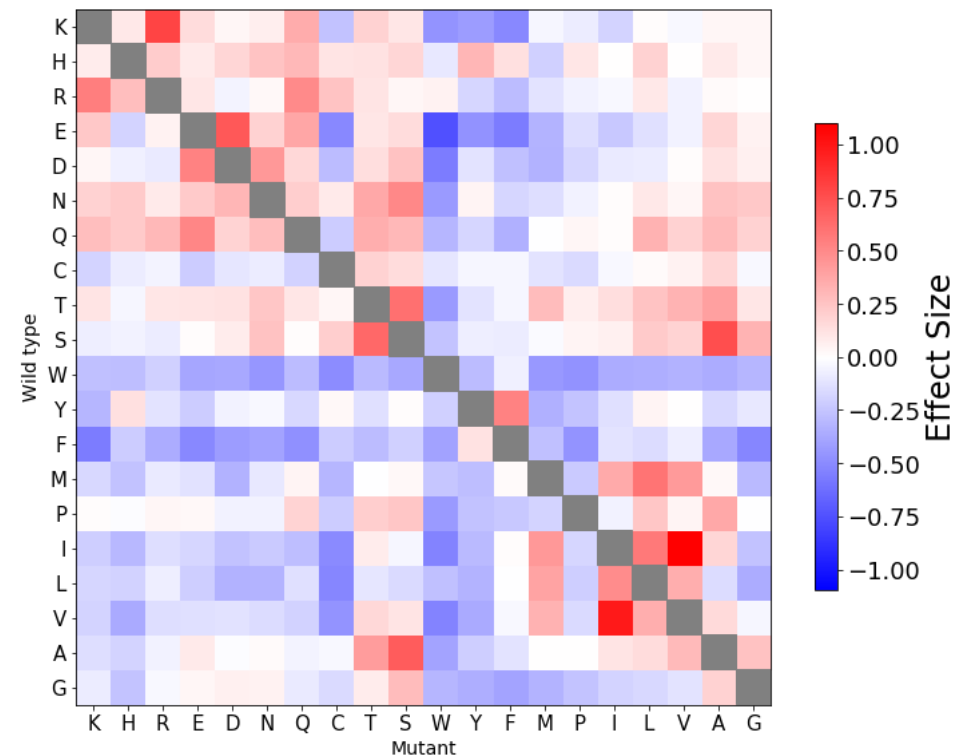


Comparing MAVE and VEP scores



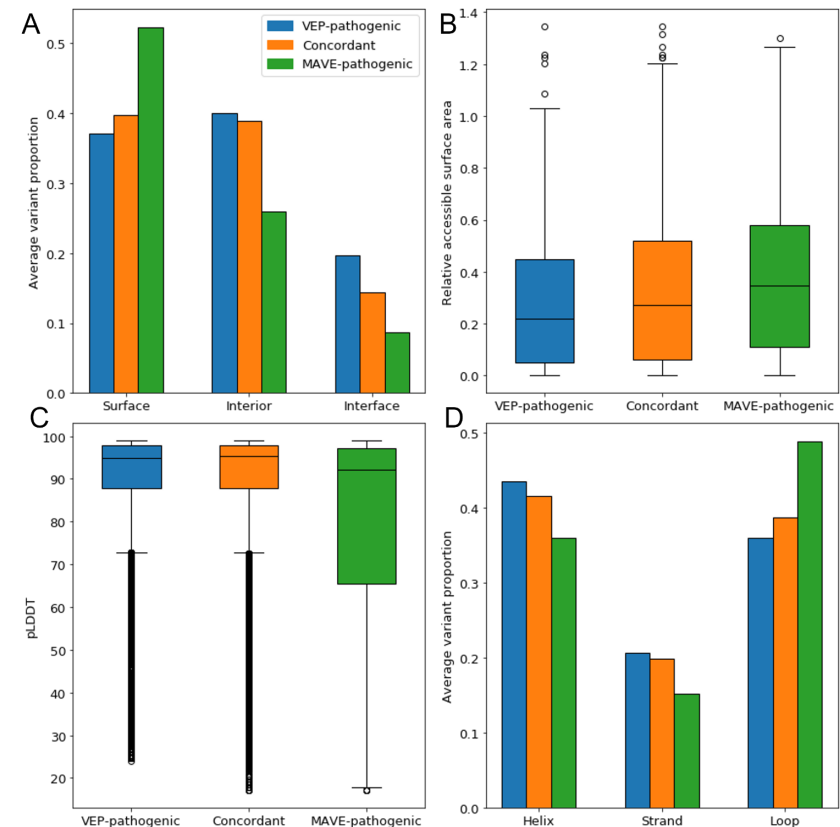
Amino acid substitution patterns

- Heatmap compares how often MAVEs vs VEPs rate each amino-acid substitution as damaging.
- Clear, non-random patterns emerge across all proteins.
- VEPs tend to score “classic disruptive” substitutions (e.g. hydrophobic $\leftrightarrow$ charged, aromatic changes) as more damaging than MAVEs.
- MAVEs tend to score “subtle” substitutions (e.g. charged $\leftrightarrow$ charged, small $\leftrightarrow$ small/hydrophobic) as more damaging than VEPs.
- These patterns are symmetric: direction of substitution (A $\rightarrow$ V vs V $\rightarrow$ A) shows similar behaviour.



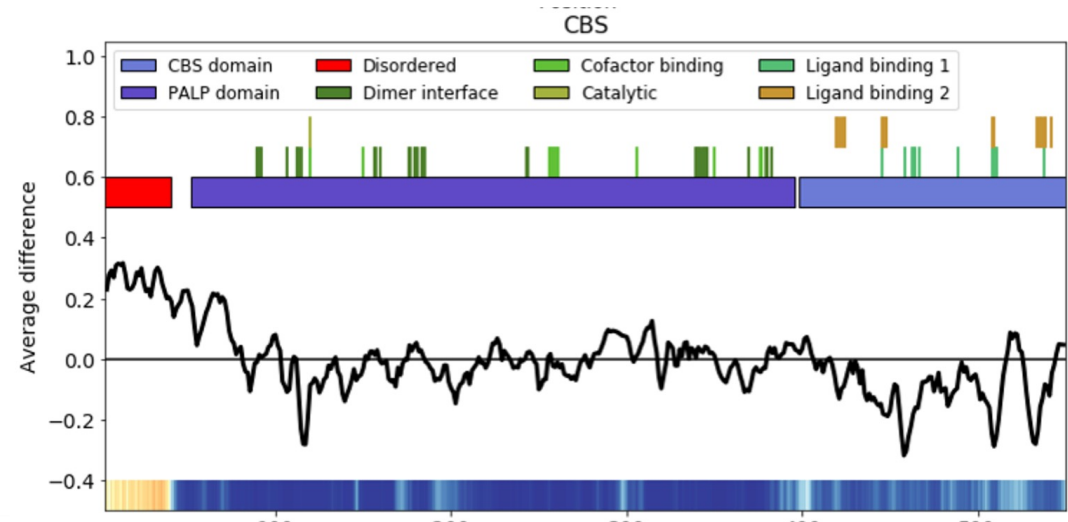
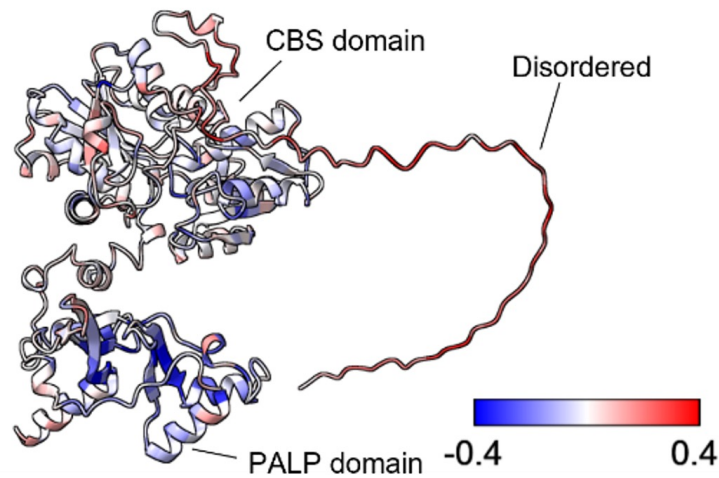
Where VEPs and MAVEs diverge

- We looked at the 5% most discordant variants in each direction.
 - **VEP-damaging:** variants VEPs rate much more damaging than MAVEs.
 - **MAVE-damaging:** variants MAVEs rate much more damaging than VEPs.
- VEP-damaging variants are enriched in buried residues and protein interfaces.
- MAVE-damaging variants are enriched on solvent-exposed surfaces and in regions with low structural confidence / intrinsic disorder.
- Secondary structure:
 - helices and strands enriched for VEP-damaging
 - loops enriched for MAVE-damaging



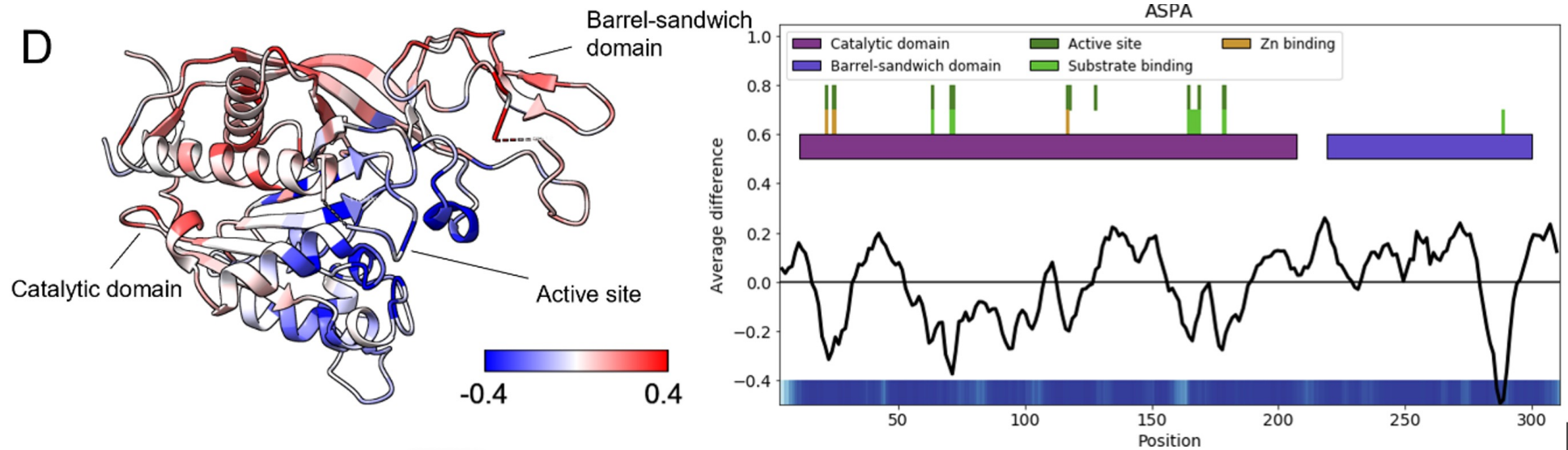
CBS

B



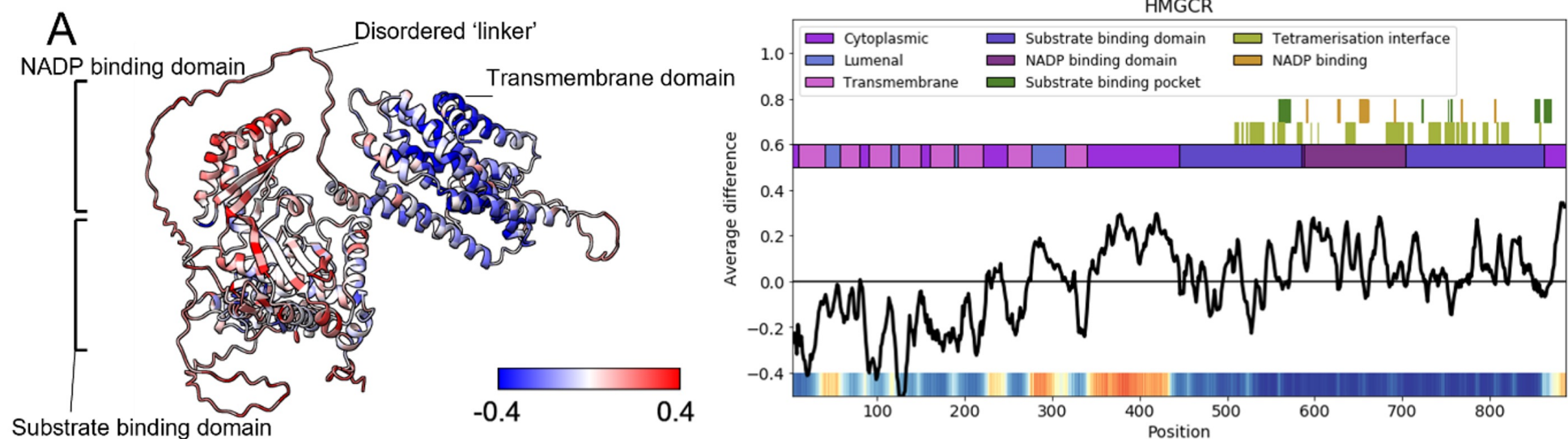
- **MAVE:** yeast complementation in *S. cerevisiae* measuring relative growth rate
- **N-terminal disordered region:** MAVEs see many variants here as functionally important, while VEPs mostly call this region benign because it is weakly conserved.
- **Central catalytic domain:** VEPs and the MAVE mostly agree on which variants are damaging here, consistent with strong conservation around the active site.
- **C-terminal regulatory domain:** VEPs flag many variants as damaging, but the yeast assay tolerates quite a lot of change, probably because regulation by the CBS domain is less critical for basic rescue of growth under the assay conditions than it is in human physiology.

ASPA



- MAVE uses VAMP-seq, which reports **protein abundance** rather than catalytic activity.
- In the active-site region, VEPs score variants as much more damaging than the MAVE does.
- Likely reason: many pathogenic ASPA variants disrupt **enzyme function** without destabilising the protein.
- VAMP-seq is blind to these catalytic defects, so their impact is underestimated.
- This is a clear example where the mechanism assayed (stability) does not match the disease mechanism (loss of catalytic activity).

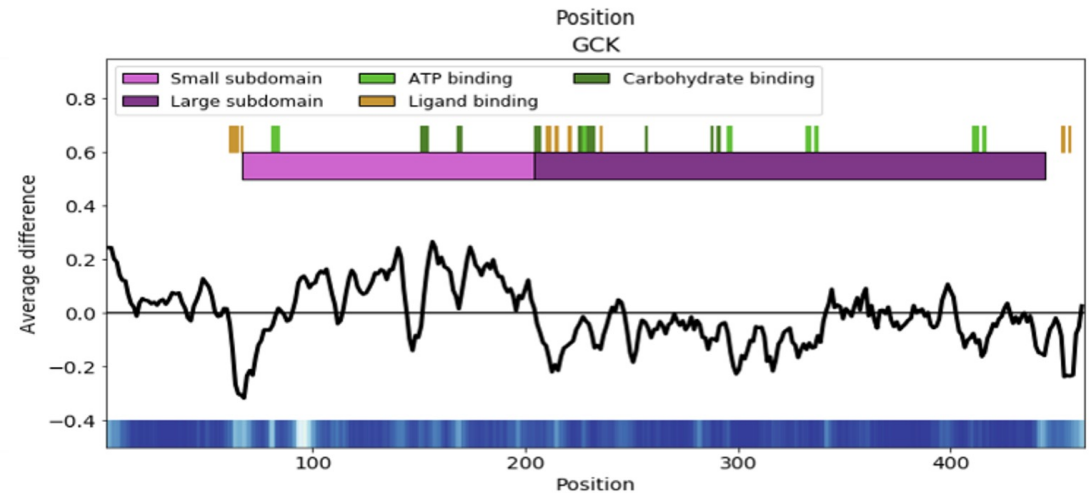
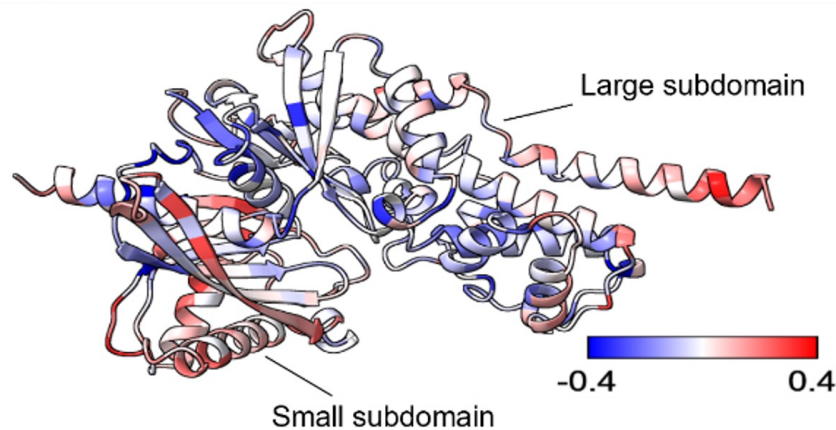
HMGCR



- **MAVE**: yeast complementation assay read out as growth, largely reflecting whether the catalytic activity is sufficient to support sterol synthesis.
- VEP-MAVE discordance is concentrated in the N-terminal sterol-sensing and transmembrane region (residues 61–218), where VEPs call variants strongly damaging but the MAVE scores are relatively mild
- TM helices are very hydrophobic and evolutionarily conserved so VEPs treat substitutions as highly damaging, but in the yeast assay, they may leave enough enzyme activity to support growth, so they seem less damaging.

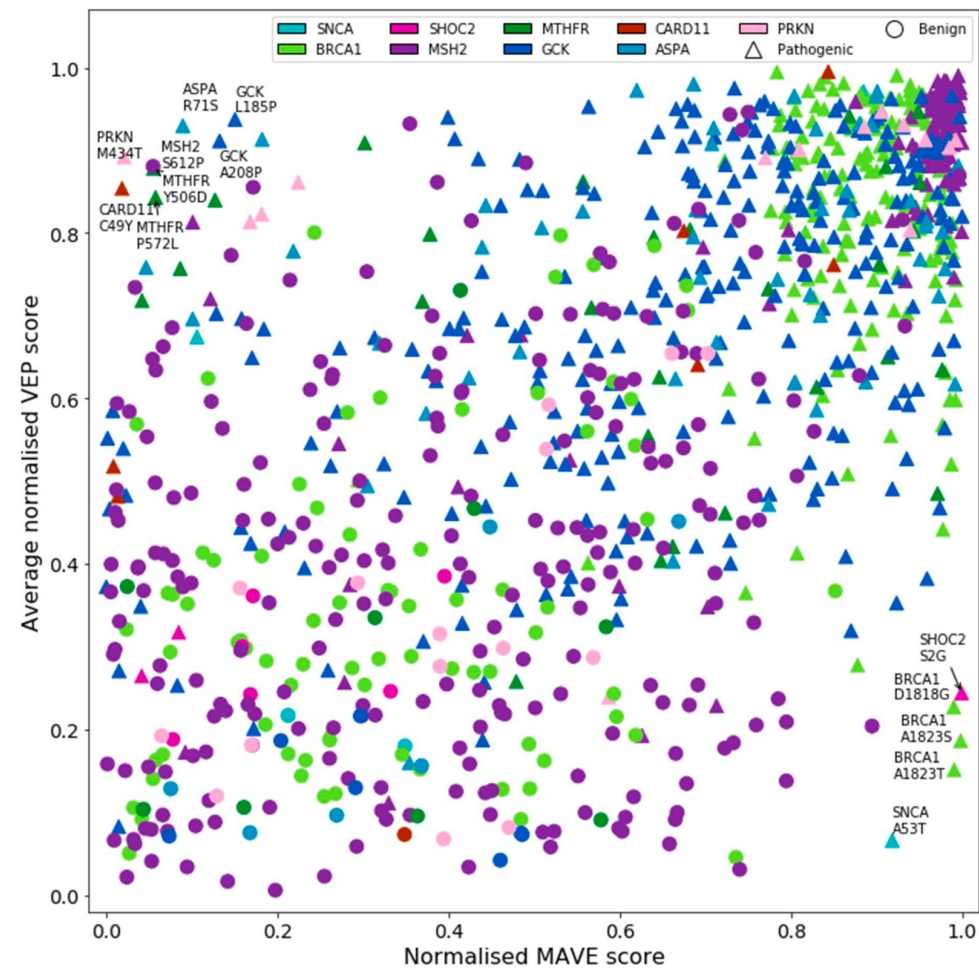
GCK

C



- The GCK MAVE uses yeast complementation: growth on glucose reports catalytic activity.
- Most of the protein shows modest VEP–MAVE differences, the small hexokinase subdomain (residues ~67–203) shows a clear positive average difference: MAVEs score more variants as damaging than VEPs.
- This region lies over the active site and is highly sensitive to changes in catalytic efficiency, which the yeast assay captures well.
- VEPs tend to undercall variants here, likely because parts of this flexible regulatory subdomain appear less constrained in sequence and structure than their functional importance implies.

MAVE vs DMS of clinical variants



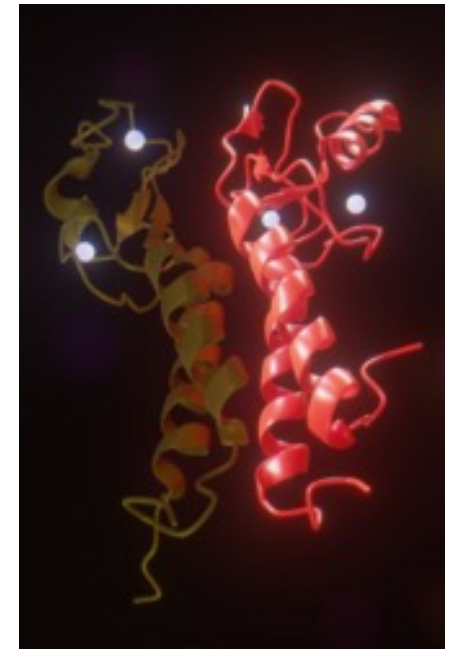
SNCA

- **MAVE assay:** Yeast toxicity assay where α -synuclein variants are expressed in *S. cerevisiae* and growth reflects aggregation toxicity.
- **Outlier variant:** A53T, a well-established pathogenic mutation causing early-onset Parkinson's disease.
- **MAVE result:** Strongly damaging. Increased aggregation reduces yeast growth, matching known biology.
- **VEP result:** Benign. Conservation-based predictors are misled because threonine is common at this position among homologues.
- **Take-home:** A clear case where evolutionary signal fails, but the phenotype-based assay captures the true pathogenic mechanism.



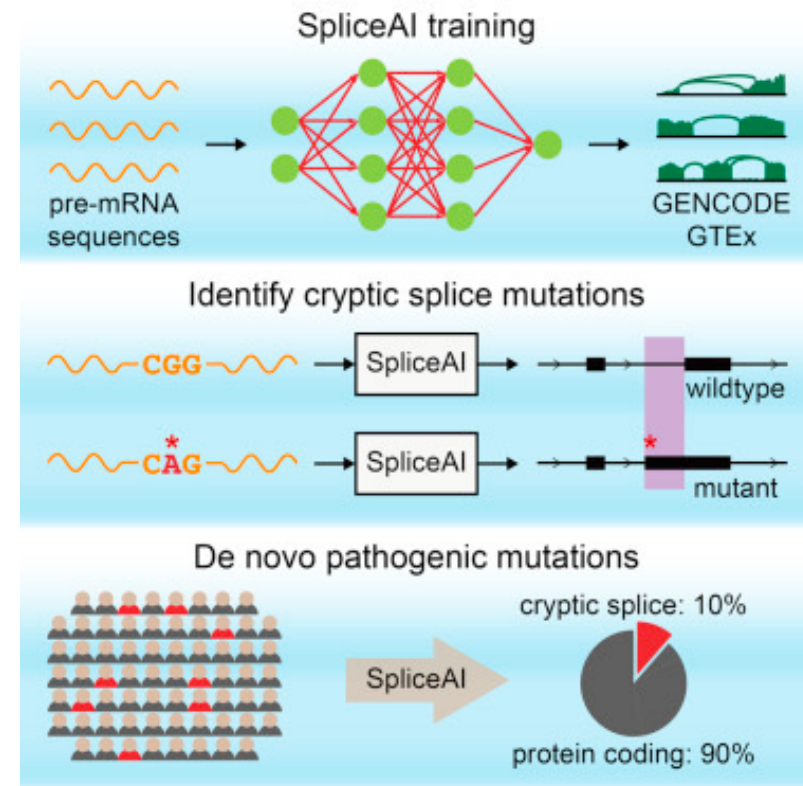
BRCA1

- **MAVE assay:** Saturation genome editing in HAP1 cells
- **Outlier variants:** A1823T, A1823S, and D1818G, all classified as pathogenic or likely pathogenic in ClinVar.
- **MAVE result:** All three are scored as damaging, consistent with their known disruption of normal BRCA1 splicing.
- **VEP result:** All are predicted as benign by top VEPs, which do not capture splice-altering effects from primary protein sequence alone.
- **Take-home:** Variants that perturb splicing can look benign to sequence/structure-based VEPs but are correctly identified in assays performed in the native genomic context.



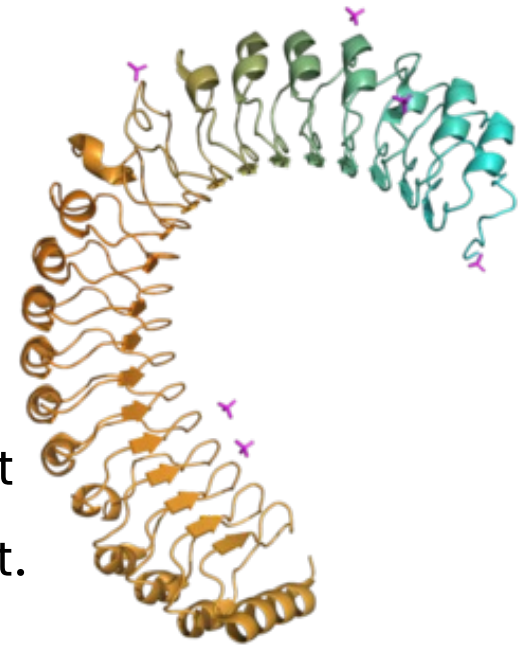
Predicting splicing impacts

- Variants that cause mis-splicing are a common cause of genetic disease
- Sometimes the impacts of missense variants are actually due to splicing effects
- SpliceAI is the most widely used splicing predictor



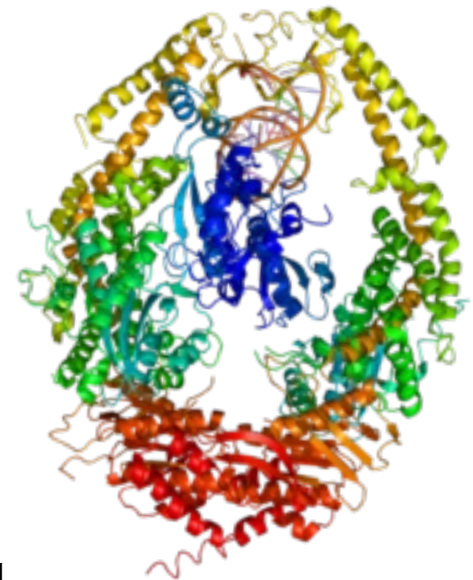
SHOC2

- **MAVE assay:** Human KRAS-mutant cancer cell line with endogenous SHOC2 knocked out and variant SHOC2 reintroduced by viral delivery; growth reflects proper MAPK pathway regulation.
- **Outlier variant:** S2G, a known pathogenic gain-of-function mutation associated with Noonan-like syndrome.
- **MAVE result:** Scored as damaging because the variant mislocalises SHOC2 through creation of an N-myristoylation site, disrupting normal signalling output.
- **VEP result:** Predicted benign, as conservation and typical structural features do not capture this rare mechanism of aberrant membrane targeting.
- **Take-home:** Unusual gain-of-function mechanisms can evade VEPs even when the cellular assay clearly detects loss of correct functional behaviour.



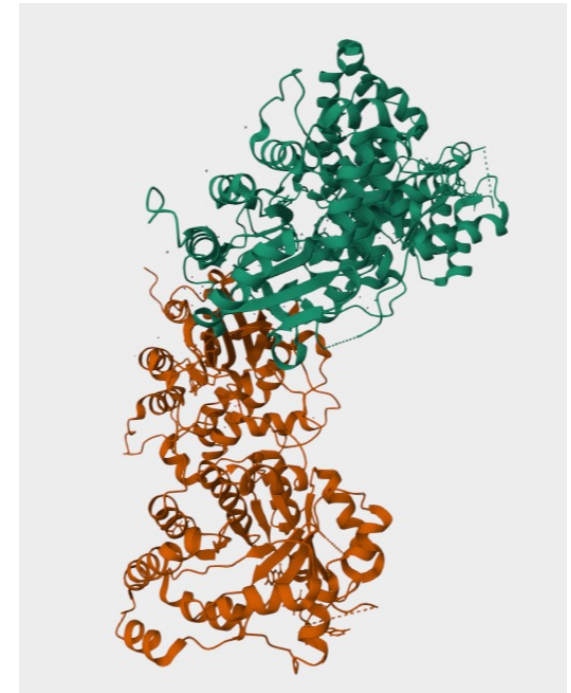
MSH2

- **MAVE assay:** Reverse-survival selection in HAP1 cells. Loss of mismatch repair increases survival in 6-TG, so loss-of-function variants appear “fitter” than wild type.
- **Outlier variant:** S612P, labelled *likely benign* in ClinVar.
- **MAVE result:** Scores as *benign* (wild-type-like), meaning the assay reports no reduction in mismatch-repair function.
- **VEP result:** Scores as *damaging* across all predictors. The position is strongly conserved, sits in an α -helix where proline is rarely tolerated, and the variant is absent from gnomAD.
- **Context at the same residue cluster:** Multiple neighbouring helix variants (A600D/V, D603Y/G/V, A604D, etc.) are *ClinVar pathogenic*, supporting that this local structural region is generally intolerant.
- **Classification complication:** The same MAVE dataset was used as functional evidence for the ClinVar benign call, raising concern the label reflects the assay rather than the biology.
- **Take-home:** This is likely an assay-driven ClinVar misclassification, not a VEP failure. The structural context and neighbouring pathogenic variants support the VEP interpretation.



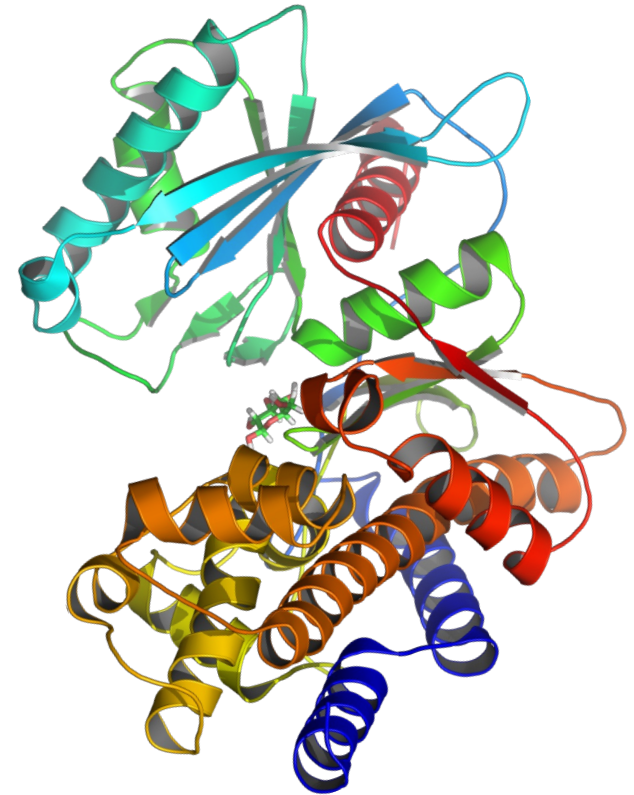
MTHFR

- **MAVE assay:** Yeast complementation of MET13. Yeast growth reflects whether human MTHFR retains enough activity to supply methionine; moderate activity can fully rescue growth.
- **Outlier variants:** Y506D and P572L, both pathogenic or likely pathogenic in ClinVar, and both supported by human biochemical evidence of **severely reduced enzyme activity**.
- **MAVE result:** Both variants score as **benign**. The yeast assay reports wild-type-like rescue even though human studies show very low residual activity (Y506D in compound heterozygotes: ~8% activity; P572L: ~3% activity).
- **VEP result:** All VEPs classify them as **damaging**, consistent with their structural context (Y506 in a dimer interface; P572 fully buried) and with the known clinical and biochemical data.
- **Interpretation:** The yeast system appears **insensitive to moderate or partial activity losses**, allowing variants with serious human metabolic impact to rescue growth. This makes the MAVE undercall these pathogenic alleles.
- **Take-home:** A case where the MAVE misses clinically meaningful loss-of-function because the complementation assay tolerates substantial reductions in catalytic efficiency, while VEPs correctly flag the variants.



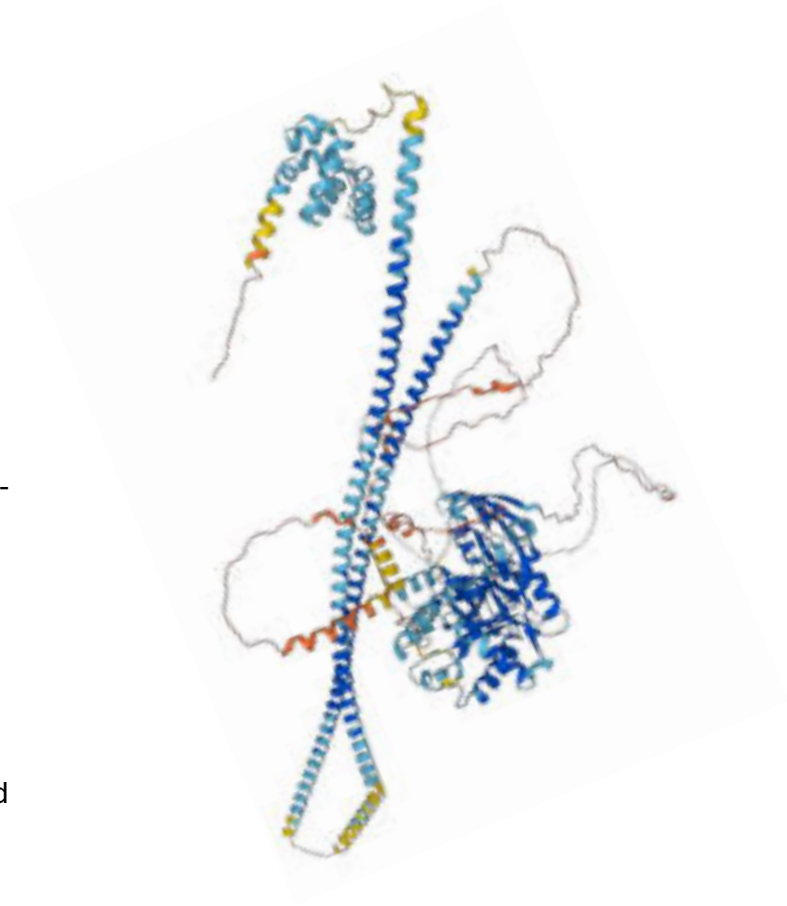
GCK

- **MAVE assay:** Yeast complementation in a strain lacking all endogenous hexokinases.
- **Outlier variants: A208P** (expert-reviewed pathogenic) and **L185P** (likely pathogenic). Both absent from gnomAD and both located in regions where most substitutions cause severe loss of function.
- **MAVE result:** Both variants score as **benign**.
- **VEP result:** All top predictors classify both variants as **damaging**, consistent with proline-induced structural disruption and local intolerance at these positions.
- **Interpretation:** These are almost certainly **MAVE false negatives**. The assay shows unusually high noise at these sites, and both variants contradict the strong local pattern where nearby substitutions are strongly deleterious.
- **Take-home:** Even when domain-level trends are sensible, individual MAVE datapoints can misclassify pathogenic variants, whereas the VEP call is more consistent with clinical and structural evidence here.



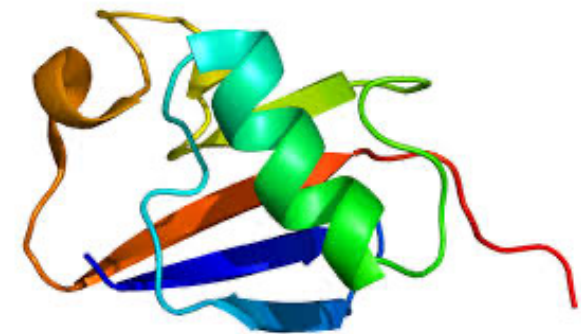
CARD11

- **MAVE assay:** Saturation genome editing in a human B-cell lymphoma line (TMD8). Growth depends on NF- κ B signalling, so **higher NF- κ B activity produces higher cellular fitness**.
- **Outlier variant: C49Y**, a known pathogenic **gain-of-function** mutation that drives excessive NF- κ B activation and causes immune dysregulation syndromes.
- **MAVE result:** Appears **benign** (or even “fitter than wild type”) because the assay reads increased NF- κ B signalling as *higher* fitness. The pathogenic gain-of-function therefore looks favourable in this context.
- **VEP result:** Predicts **damaging**, consistent with the known biology and the established pathogenic classification.
- **Interpretation:** This is not a MAVE failure: the assay is correctly detecting increased signalling, but the **direction of the phenotype is inverted** relative to human disease. Pathogenic gain-of-function variants can look benign unless the scores are rescaled (“flipped”).
- **Take-home:** When the assay maps pathogenic gain-of-function to increased growth, raw MAVE scores can mislead, while VEPs retain the correct directional interpretation.



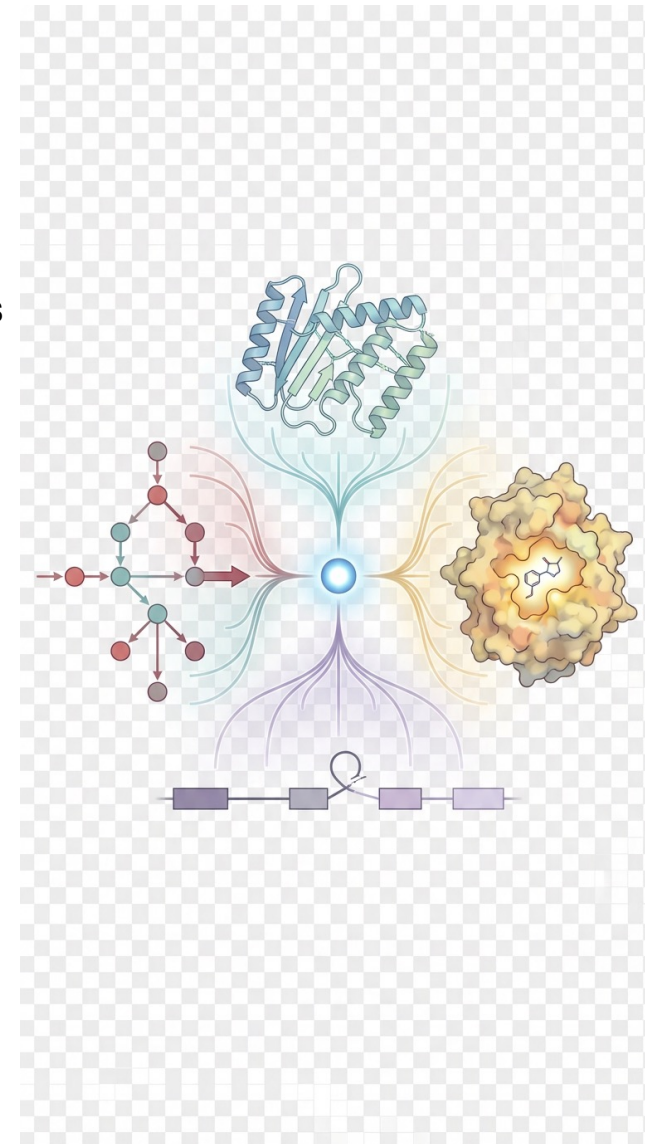
ASPA/PRKN

- **MAVE assay:** VAMP-seq in human cells, where variant abundance reports on **protein stability** rather than catalytic or enzymatic function.
- **Outlier variants:** **ASPA R71S**, pathogenic, located in the active site pocket; **PRKN M434T**, pathogenic, near the catalytic centre of the E3 ligase domain.
- **MAVE result:** Both variants score as **benign**, because their protein abundance is close to wild type. VAMP-seq does not detect catalytic or binding defects when stability is intact.
- **VEP result:** All major predictors classify both variants as damaging, matching structural expectations and clinical evidence.
- **Interpretation:** These are classic **catalytic-loss-of-function** variants with preserved folding. The stability-focused MAVE misses their true molecular impact, while VEPs retain the correct interpretation.
- **Take-home:** Stability-only assays systematically undercall variants that disrupt enzymatic activity without affecting abundance.

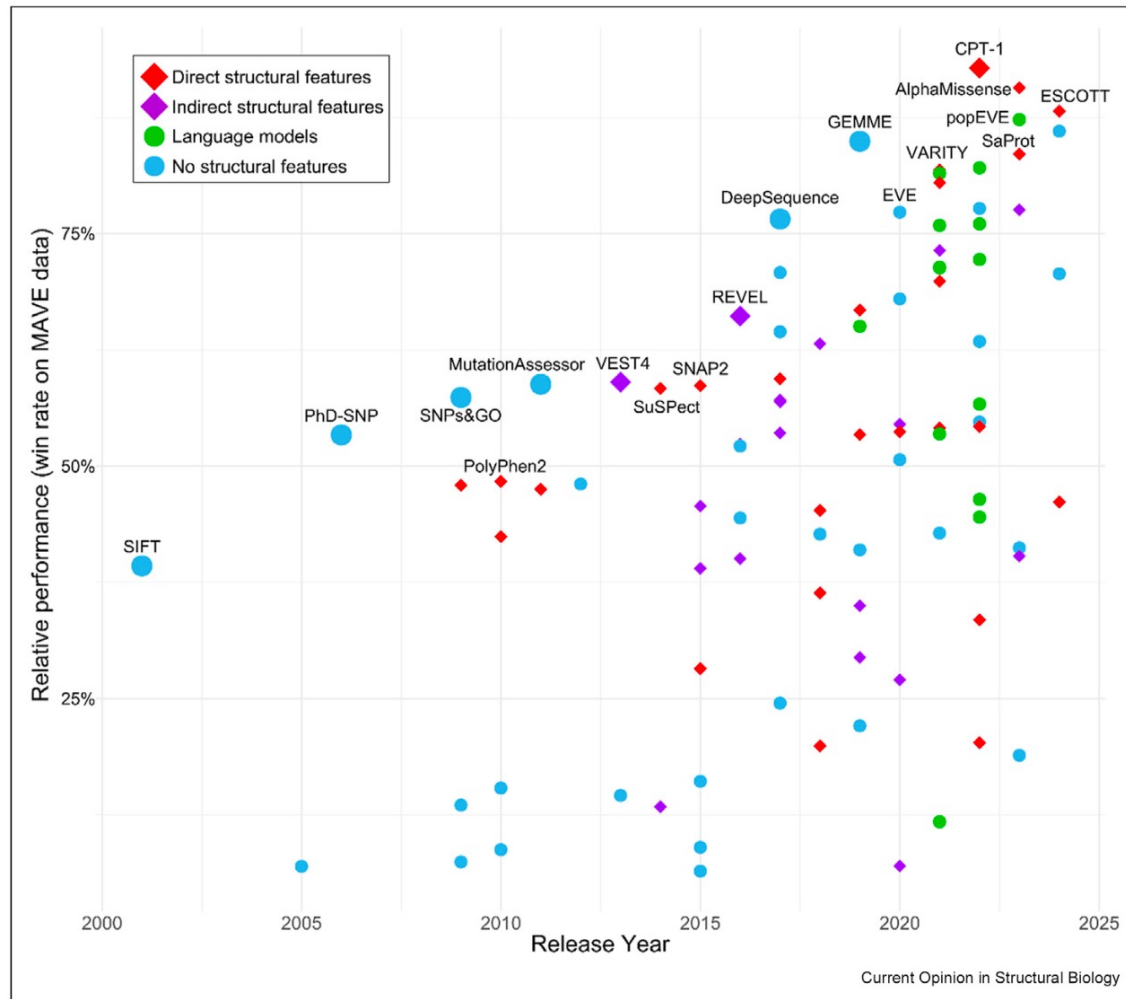


The importance of mechanisms

- The outlier variants show that disagreement between VEPs and MAVEs is mostly **not random**.
- Most large discrepancies come from **mechanistic blind spots** in one approach or the other.
- **MAVEs** can miss:
 - catalytic or binding defects when the assay only measures stability
 - gain-of-function variants when increased activity boosts growth
 - human-specific regulatory mechanisms not captured in yeast
- **VEPs** can miss:
 - splice-altering or expression-altering variants
 - unusual gain-of-function mechanisms or toxic effects that do not follow evolutionary constraint
- **Take-home:** Many disagreements stem from *which* molecular mechanisms each method can or cannot detect.
- *Can we predict molecular disease mechanisms? To understand mechanisms, we need to consider structure...*



Recent best methods use structures (*but still rely mostly on evolution*)



Most VEPs tell us nothing about *mechanisms*



Current Opinion in Structural Biology

Volume 92, June 2025, 103023

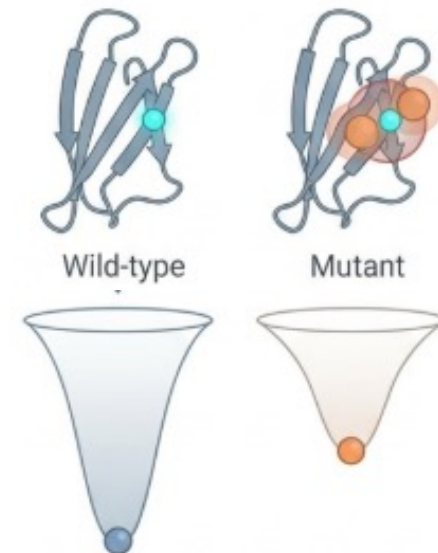


Leveraging protein structural information to improve variant effect prediction

Lukas Gerasimavicius¹, Sarah A. Teichmann^{2,3}, Joseph A. Marsh¹

Stability ($\Delta\Delta G$) Predictors

- A class of VEPs that estimate how a missense change alters **protein stability** ($\Delta\Delta G$) using **protein structure models**
- **Physics/energy-function methods**
 - Use empirical force fields, rotamer libraries, and side-chain packing energy terms.
 - Examples include FoldX, Rosetta-based $\Delta\Delta G$ protocols, and related tools.
 - Fast, interpretable, and widely used for screening variants on known structures.
- **Modern ML + structure models**
 - Machine-learning frameworks trained on curated experimental $\Delta\Delta G$ datasets.
 - Use structural features (solvent accessibility, packing density, hydrogen bonding, local geometry).
- **What they provide**
 - A mechanistic predictor specifically targeting **folding stability** and, by extension, abundance.

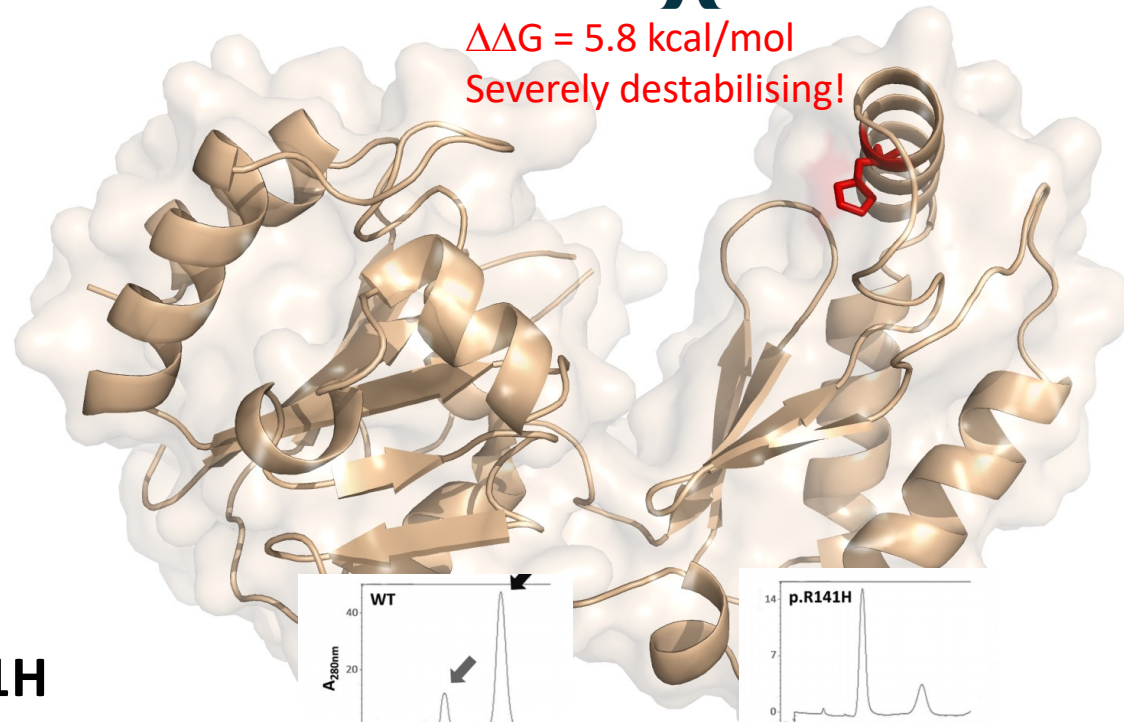


Protein stability predictors estimate how mutations affect protein folding stability ($\Delta\Delta G$)

FoldX

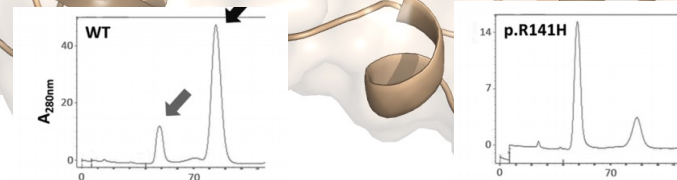
$\Delta\Delta G = 5.8 \text{ kcal/mol}$

Severely destabilising!



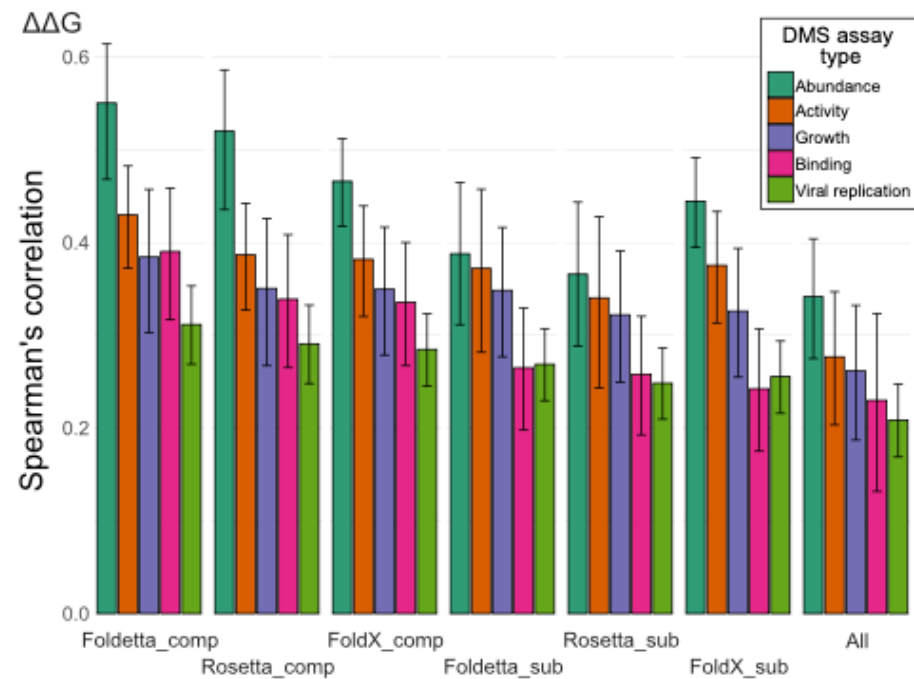
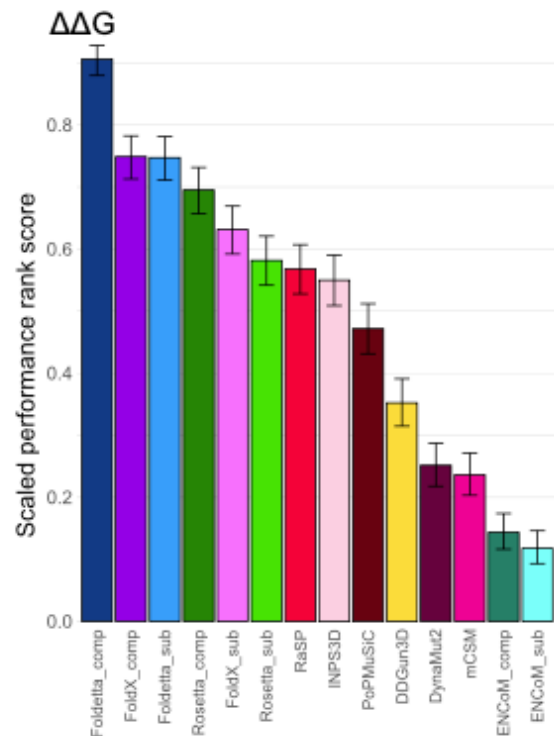
***PMM2* R141H**

0.4% allele frequency



Yuste-Checa et al (2015) Hum Mut

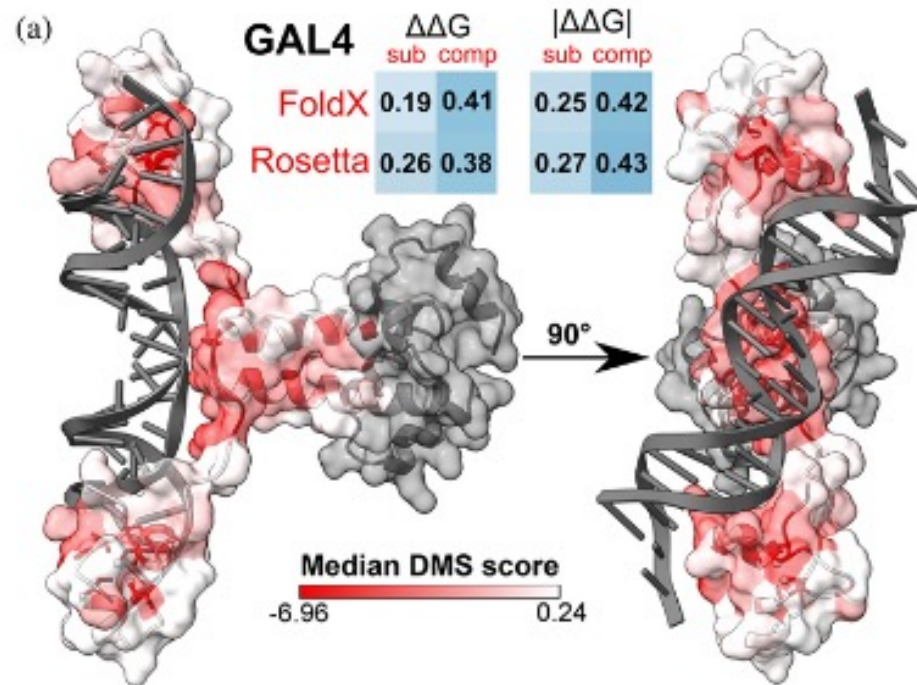
Benchmarking stability predictors with MAVE data



- Long-standing methods FoldX and Rosetta do best
- Unsurprisingly, correlation his highest with abundance-based MAVEs (e.g. VAMP-seq)

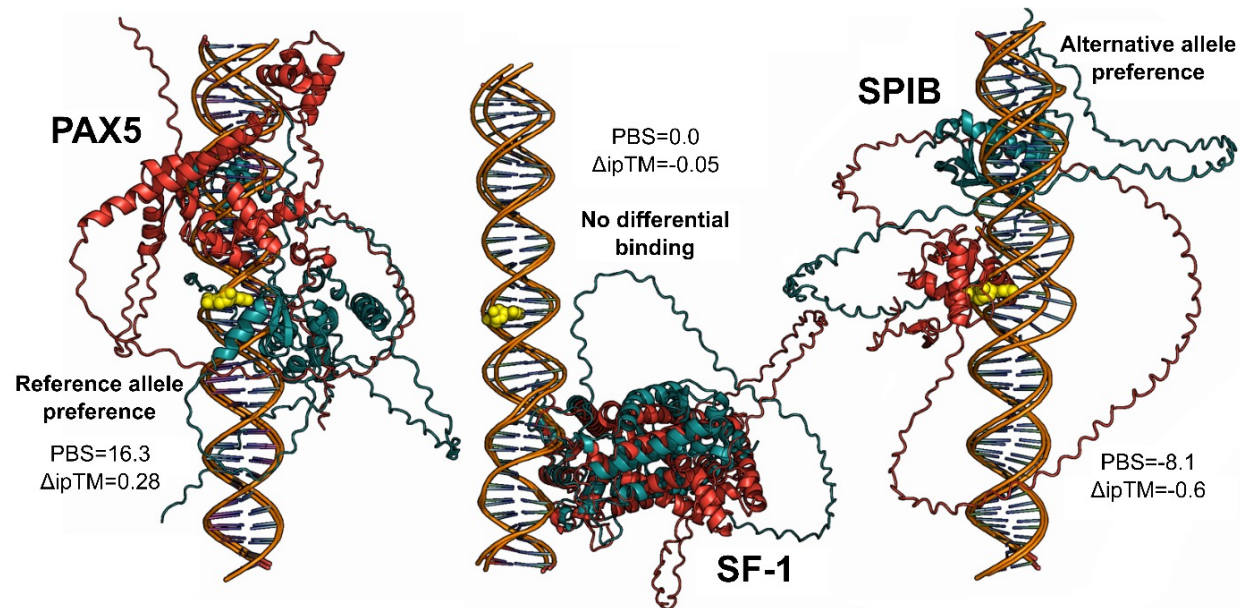
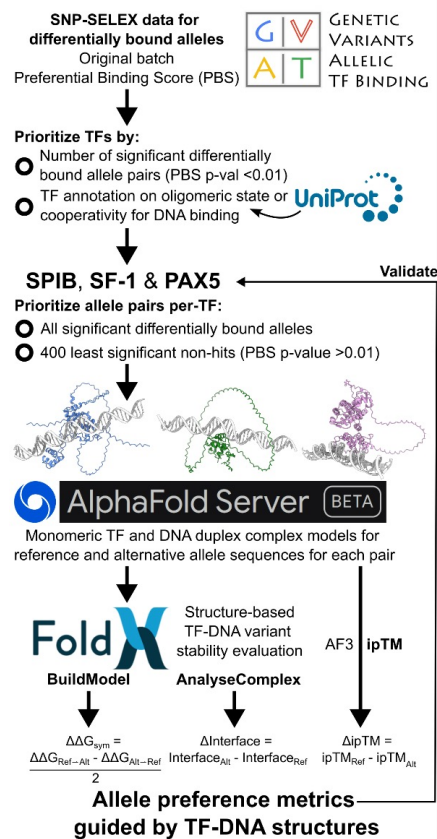
Gerasimavicius et al
(2023) Protein Sci

Some stability predictors also work on complex structures



- Correlation with MAVE data is greatly improved when full DNA-bound complex is considered

FoldX can also work for non-coding variants



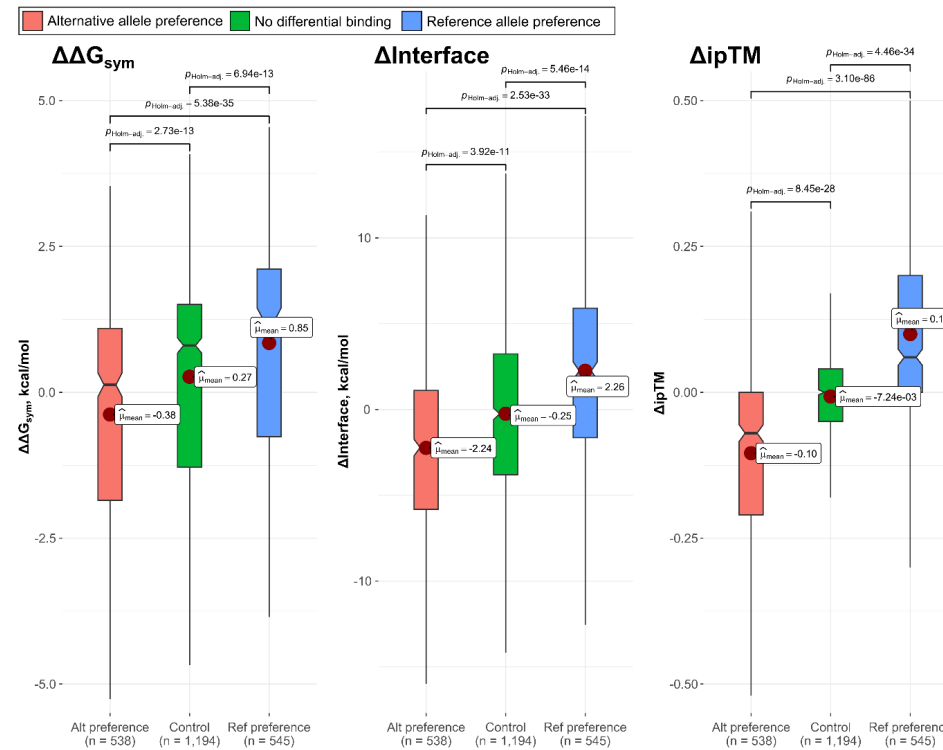
A structure-guided approach to non-coding variant evaluation for transcription factor binding using AlphaFold 3

Lukas Gerasimavicius, Simon C. Biddie, Joseph A. Marsh

doi: <https://doi.org/10.1101/2025.03.20.644433>

Accepted at Nucleic Acids Research

AF3 models show strong agreement with allele binding preference



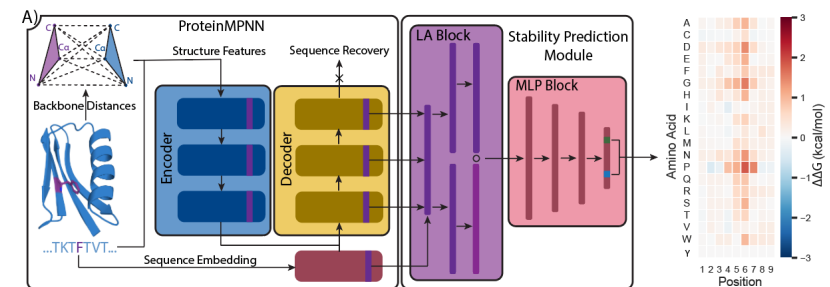
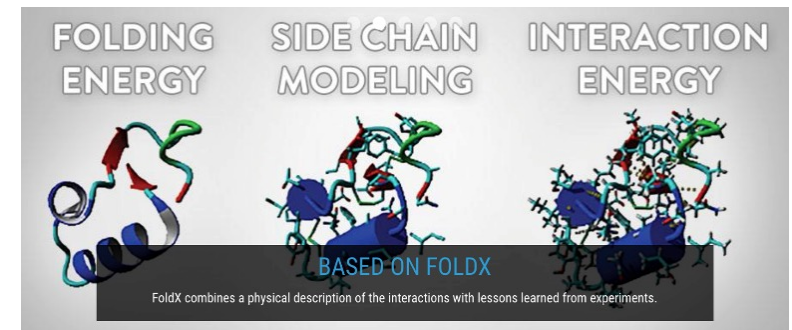
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doi: <https://doi.org/10.1101/2025.03.20.644433>

Which stability predictors to use

- **FoldX** - despite being an older method, performs very well and can also be applied to complexes and non-coding variants
 - Command line, but human AlphaFold-based predictions are available in **ProtVar**
- **ThermoMPNN** – state-of-the-art neural network-based model, not included in our benchmarks, but very promising performance
 - Performance on complexes uncertain



Missense3D Imperial College London

Home Documentation Dataset Contact

MISSENSE 3D

Missense3D predicts the structural changes introduced by an amino acid substitution and is applicable to analyse both PDB coordinates and homology-predicted structures.

Please cite: Ittisoponpisan, S., Islam, S.A., Khanna, T., Alhuzimi, E., David, A. & Sternberg, M.J.E. (2019) Can Predicted Protein 3D Structures Provide Reliable Insights into whether Missense Variants Are Disease Associated? J. Mol. Biol. 431, 2197-2212.
<https://doi.org/10.1016/j.jmb.2019.04.009>

Variant information

PDB code	3sc0
OR upload here (5MB limit)	Choose File No file chosen
PDB chain ID	A
Residue ID	155
Wild-type amino acid (Required just as a check)	Gly (G Glycine)
Mutant amino acid	Glu (E Glutamic acid)
Email (under construction)	Email

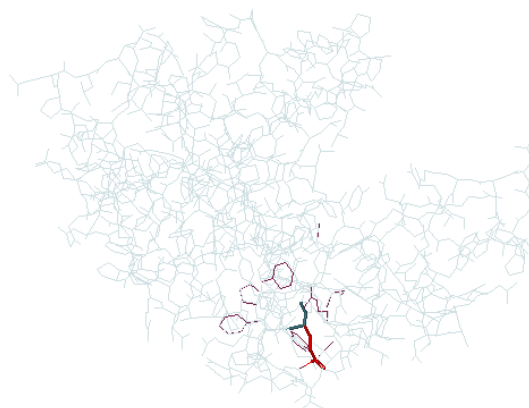
[Reset](#) [Back to Home](#)

SUBMIT

Has an option to include the Uniprot sequence ID, in case the residue numbering doesn't match the structure

Download your results here [Download](#)

IMPORTANT - Your results will be deleted from our server after 24-hours. Therefore please download the zip file above.



3sc0 (A)

Residue ID: 155

Variant: GLY > GLU

Structural damage detected

Buried charge introduced Buried Gly replaced

Display control

[Export image](#)

Model ☒ both ☐ wild type ☐ mutant

Chain colour wild type mutant

Residue colour wild type mutant

Centre ☒ protein ☐ variant

Show as

Surface ☐ on ☒ off

Label ☒ none ☐ local

Image quality ☒ low ☐ high

(For advanced user: [console](#) [info](#) [show](#) / [hide](#))

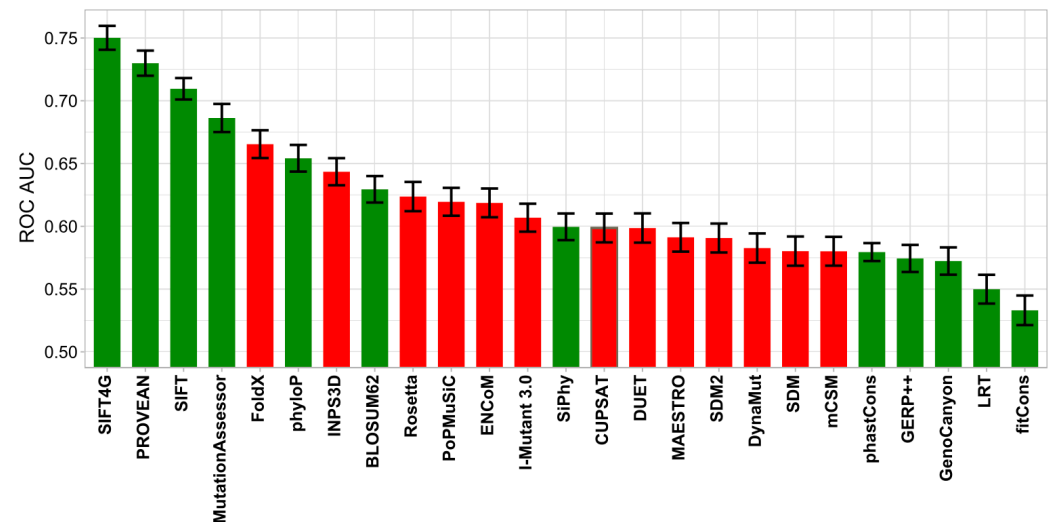
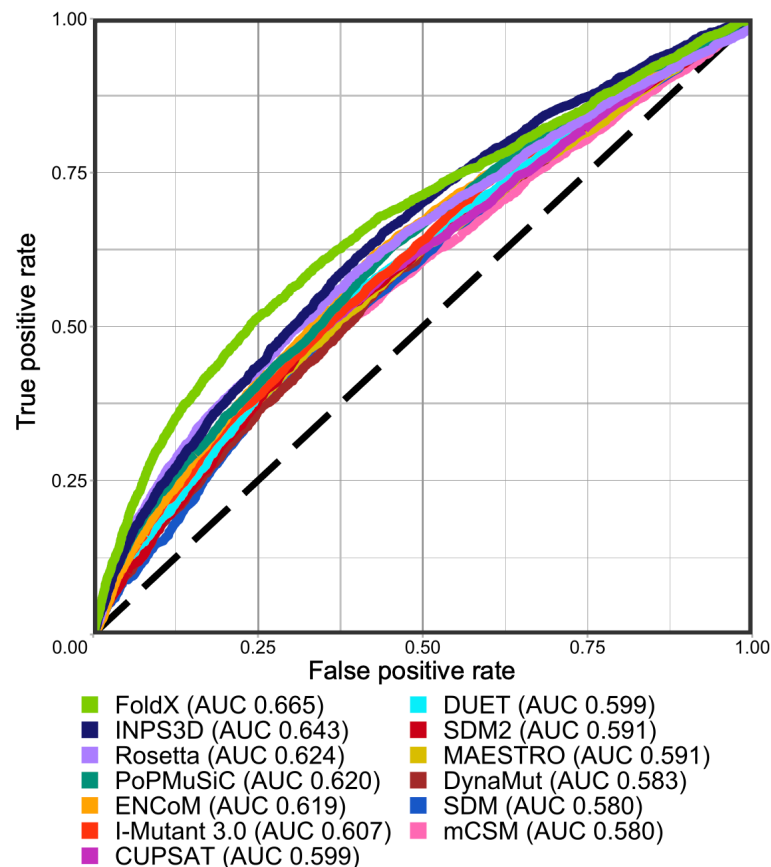
[Download mutant PDB structure](#)

Detailed analysis

Click on each feature for further information

Disulphide breakage The wild-type residue is not CYS so it cannot form a disulphide bond. Criterion	Buried Pro introduced This substitution does not introduce a proline. Criterion	Clash This substitution does not trigger clash alert. The local clash score for wild type is 16.65 and the local clash score for mutant is 34.39. Criterion
Buried hydrophilic introduced This substitution does not replace a buried hydrophobic residue with a hydrophilic residue. The wild-type residue GLY is buried neutral with RSA 0.0% and the mutant residue GLU is buried hydrophilic with RSA 0.5%. Criterion	Buried charge introduced ⓘ This substitution replaces a buried uncharged residue (GLY, RSA 0.0%) with a charged residue (GLU). Criterion	Secondary structure altered This substitution does not alter the secondary structure 'B' (isolated β-bridge). Criterion
Buried charge switch This substitution does not trigger buried charge switch alert. The wild-type residue GLY is buried uncharged with RSA 0.0% and the mutant residue GLU is buried negative-charged with RSA 0.5%. Criterion	Disallowed phi/psi This substitution does not trigger disallowed phi/psi alert. The phi/psi angles are in favored region for wild-type residue and in allowed region for mutant residue. Criterion	Buried charge replaced This substitution does not replace a buried charged residue with an uncharged residue. The wild-type residue GLY is buried uncharged with RSA 0.0% and the mutant residue GLU is buried charged with RSA 0.5%. Criterion
Buried Gly replaced ⓘ This substitution replaces a buried GLY residue (RSA 0.0%) with a buried GLU residue (RSA 0.5%). Criterion	Buried H-bond breakage The buried wild-type residue is not involved in any side-chain / side-chain H-bond(s) and/or side-chain / main-chain bond(s) H-bonds. Check H-bond Criterion	Buried salt bridge breakage The wild-type residue does not form any salt bridge. Criterion
Cavity altered The substitution leads to the expansion of cavity volume by 1.944 Å³. Criterion	Buried / exposed switch The wild-type residue GLY is buried (RSA 0.0%) and the mutant residue GLU is buried (RSA 0.5%). Criterion	Cis pro replaced Wild-type residue is not a cis proline. Criterion
Gly in a bend The wild-type residue is GLY but it is not located in a bend curvature. Criterion		

Stability predictors are poor at identifying pathogenic variants



Article | [Open access](#) | Published: 21 September 2020

Identification of pathogenic missense mutations using protein stability predictors

[Lukas Gerasimavicius](#), [Xin Liu](#) & [Joseph A. Marsh](#)

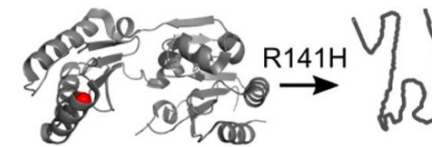
[Scientific Reports](#) **10**, Article number: 15387 (2020) | [Cite this article](#)

Nearly all pathogenic coding mutations can be split into 3 groups

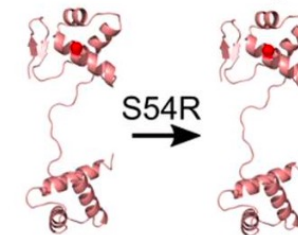
1. Loss of function – *the phenotypic effect is due to a partial or complete loss of the proteins normal biological function*

2. Gain of function – *the phenotypic effect is due to the mutant protein doing something different than wild type*

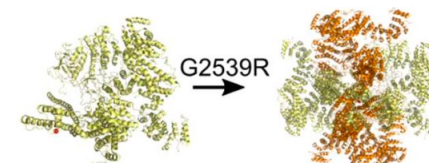
3. Dominant negative – *the phenotypic effect is due to the mutant protein blocking the normal biological function of the wild-type protein*



disrupts protein folding



alters DNA-binding specificity



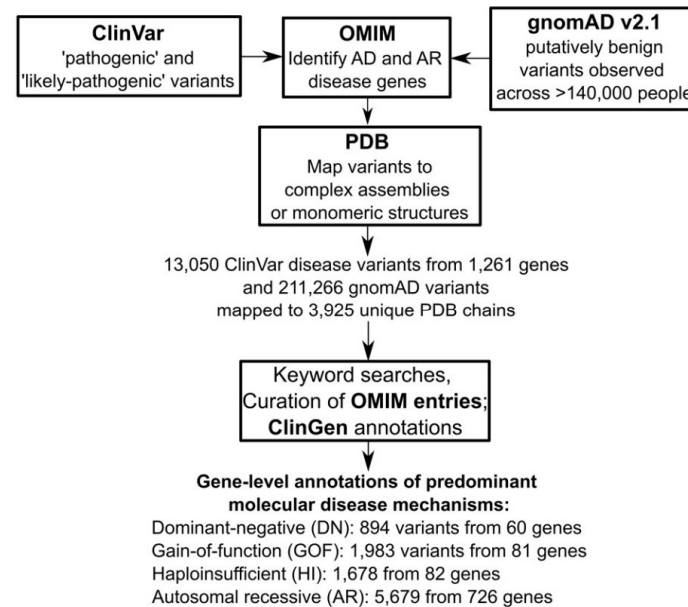
mutant assembles with wild type disrupting channel activity

Review > [Annu Rev Genomics Hum Genet.](#) 2022 Aug 31;23:475-498.
doi: 10.1146/annurev-genom-111221-103208. Epub 2022 Apr 8.

Diverse Molecular Mechanisms Underlying Pathogenic Protein Mutations: Beyond the Loss-of-Function Paradigm

Lisa Backwell ¹, Joseph A Marsh ¹

Classification of dominant mutation mechanisms



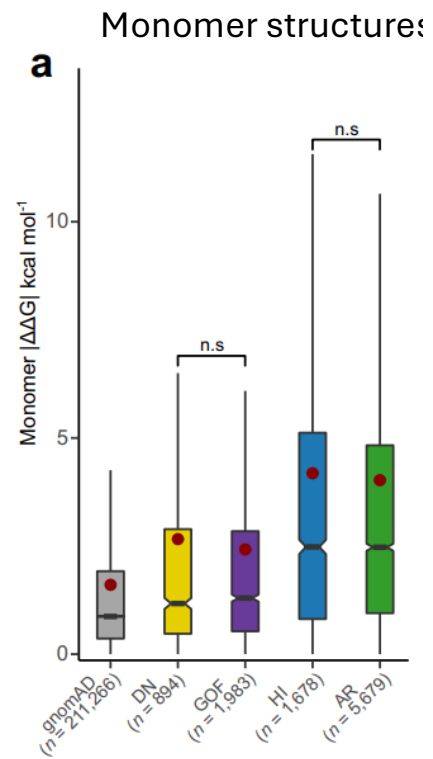
Lukas Gerasimavicius

NATURE COMMUNICATIONS | (2022)13:3895 | <https://doi.org/10.1038/s41467-022-31686-6> |

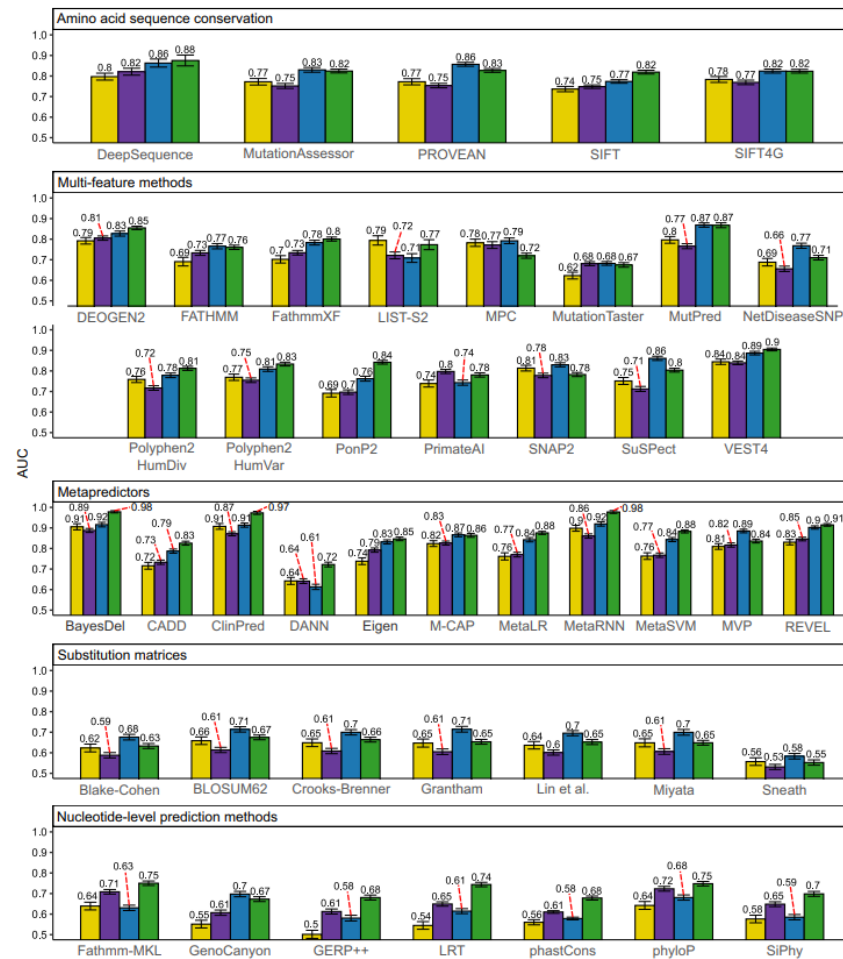
Loss-of-function, gain-of-function and dominant-negative mutations have profoundly different effects on protein structure

Lukas Gerasimavicius¹, Benjamin J. Livesey¹ & Joseph A. Marsh^{1✉}

Dominant-negative and gain-of-function mutations are milder than dominant loss-of-function mutations



DN and GOF variants are poorly predicted by VEPs



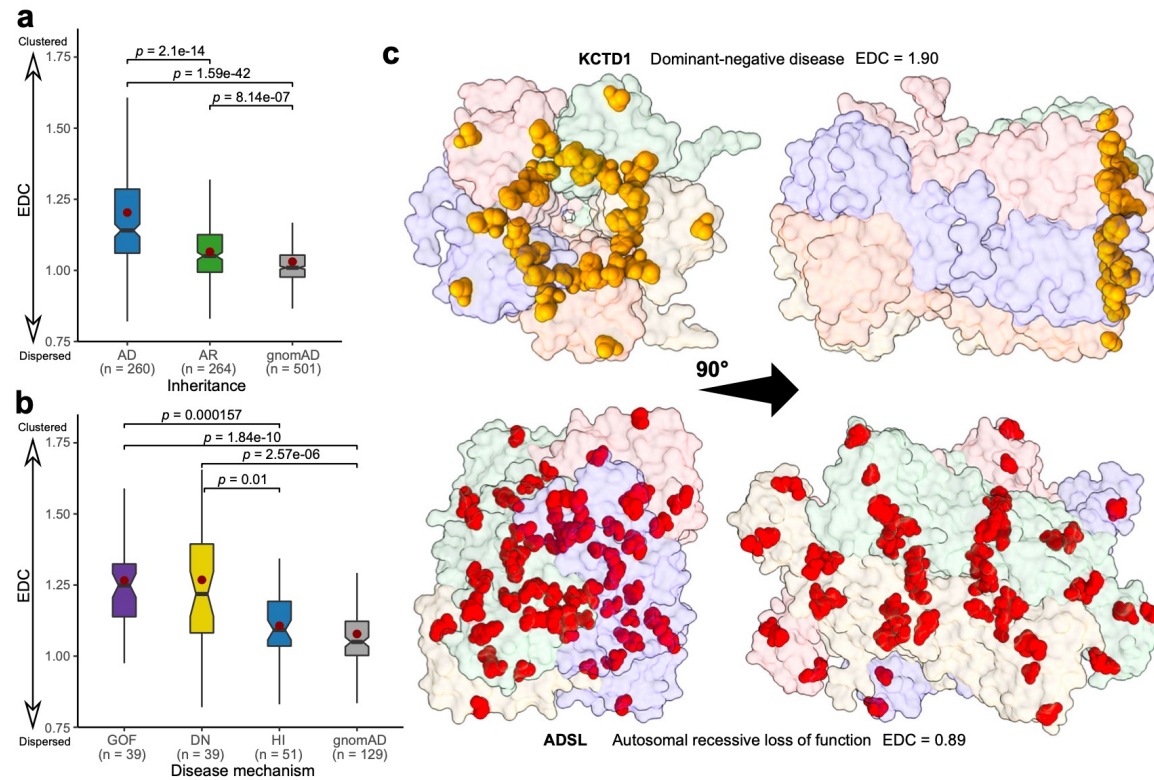
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Loss-of-function, gain-of-function and dominant-negative mutations have profoundly different effects on protein structure

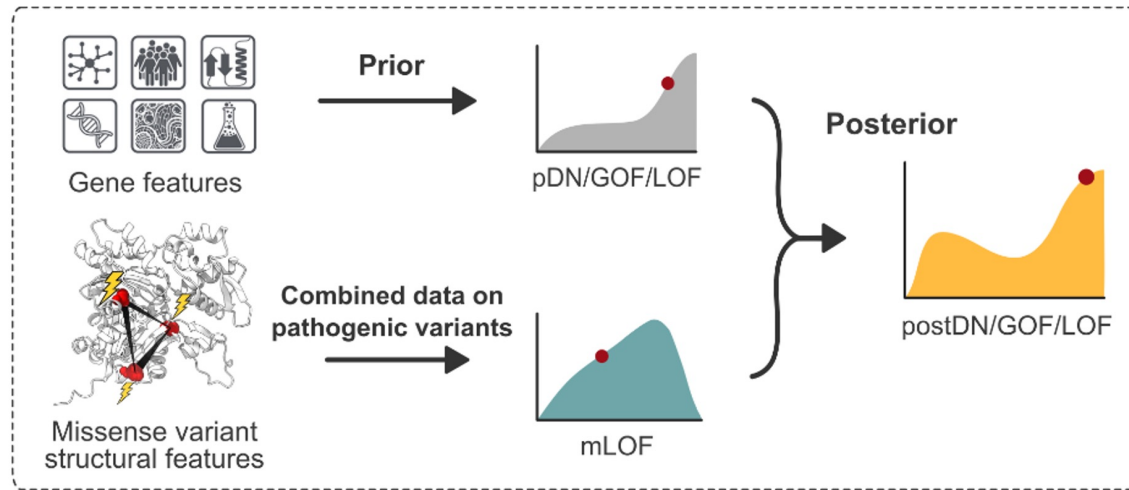
Lukas Gerasimavicius¹, Benjamin J. Livesey¹ & Joseph A. Marsh¹

DN and GOF variants cluster in 3D space

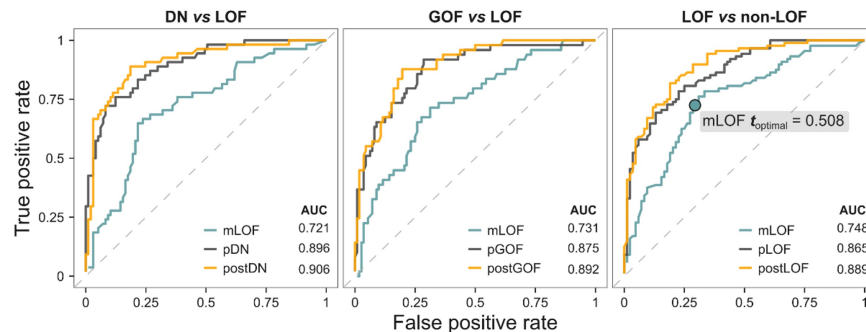
EDC =
Extent of
Disease
Clustering



Combine gene- and variant-level properties to predict molecular mechanisms



Single-phenotype genes



Article | [Open access](#) | Published: 25 September 2025

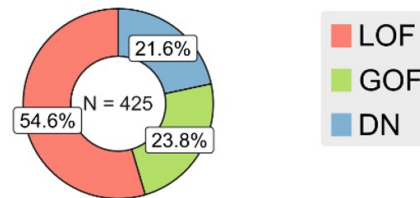
Prevalence of loss-of-function, gain-of-function and dominant-negative mechanisms across genetic disease phenotypes

Mihaly Badonyi & Joseph A. Marsh

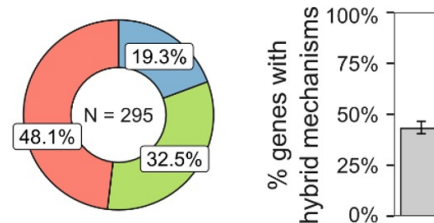
Nature Communications 16, Article number: 8392 (2025) | [Cite this article](#)

Prevalence of different mechanisms in human genetic disorders

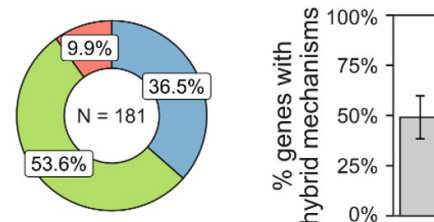
Single-phenotype, exclusively AD genes



Multi-phenotype, exclusively AD genes



AD phenotypes in AD/AR_{mixed} genes



<https://github.com/badonyi/mechanism-prediction>



Important, this is only meaningful for dominant genes!

Specify a human gene

Gene name

gene:

or UniProt accession number

id:

Input variants

Example: A108S, L730H, T454P

variants:

[Show code](#)

INFO Using the AlphaFold model (<https://alphafold.ebi.ac.uk/entry/Q15858>)
 WARNING Residue(s) 660, 834, 1143, 1959 are below the pLDDT cutoff for EDC calculation.
 INFO Calculating EDC... DONE
 INFO Retrieving $\Delta\Delta\text{Grank}$ scores... DONE

Structure-based metrics

EDC	1.465
$\Delta\Delta\text{Grank}$	0.579
mLOF	0.397

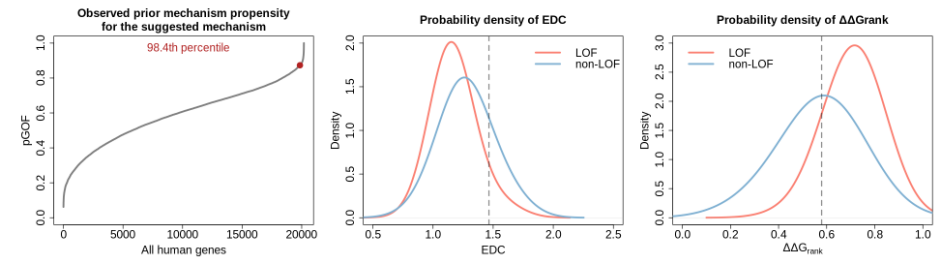
Prior mechanism propensities

pLOF	0.188
pGOF	0.873
pDN	0.834

Ranked mechanism-specific posteriors

postGOF	0.913
postDN	0.885
postLOF	0.132

VERDICT A gain-of-function mechanism is most likely for the variants.



GCH1 14:54842008-54902826

Reverse strand gene: GTP cyclohydrolase 1

Formerly known as: **DYT14, DYT5, GCH**

Also known as: **GTPCH1, DYT5a, ENSG00000131979**

Function: "Positively regulates nitric oxide synthesis in umbilical vein endothelial cells (HUVECs). May be involved in dopamine synthesis. May modify pain sensitivit Isoform GCH-1 is the functional enzyme, the potential function of the enzymatically inactive isoforms ..." [Show more >>](#) Source: [UniProt](#)

DECIPHER holds 1 sequence variants in this gene, in 1 open-access patients

Overview	Matching patient variants 34	Matching DDD research variants 0	Phenotypes	Phenotype browser	Transcripts 6	Browse
Clinical	Protein / Genomic					

Gene/disease association

Gene2Phenotype ?

- Monoallelic - Uncertain: Dystonia Type 5 **DD, Definitive**
- Biallelic - Uncertain: GTP Cyclohydrolase 1 Deficiency **DD, Definitive**

OMIM ?

600225

Morbid ?

- [Dystonia, DOPA-responsive](#) (Autosomal dominant, Autosomal recessive)
- [Hyperphenylalaninemia, BH4-deficient, B](#) (Autosomal recessive)

GeneReviews ?

- [GTP Cyclohydrolase 1-Deficient Dopa-Responsive Dystonia](#)
- [Hereditary Dystonia](#)

Gene predictive scores

pLI	0.36	?
LOEUF	0.90	?
sHet	0.157	?
pHaplo	0.93	?
pTripo	0.99	?
Missense Z Score	0.63	?

Protein predictive scores

pDN	0.458	?
pGOF	0.324	?
pLOF	0.668	? Literature support

Practical

1. Pick a MAVE dataset and compare it to VEPs for the same gene
How do they compare across the protein, and how do they compare on pathogenic and benign variants
2. Compare per-residue MAVE and VEP scores across length and mapped onto protein structure
3. Investigate pathogenic and benign outlier variants
4. Try *mechanism-prediction* (for dominant genes)

Links and datasets

<https://sites.google.com/view/mave2025-lectures>